IILM UNIVERSITY, GREATER NOIDA

School of Computer Science and Engineering | Summer Internship 2026-27

INTERNSHIP PRESENTATION

Cloud Infrastructure Automation & Containerized Deployment Platform

AWS • Terraform • Docker • Kubernetes • Jenkins • Prometheus

STUDENT DETAILS

Student Name: PawanDubey
Enrollment / Roll No: CS-2341492
Program & Sem: B.Tech CSE (4CSE8), 7th Sem

ORGANIZATION DETAILS

Organization Name: SystemaOps (systemaops.com)
Internship Role: Cloud & DevOps Engineering Intern
Duration: Summer 2026 (3 Months)

Executive Summary & Internship Overview

Cloud Infrastructure

Provisioned Multi-AZ AWS architecture using HashiCorp Terraform modules for automated VPC, ALB, EC2, and RDS deployment.

CI/CD & Containers

Automated end-to-end testing, Docker multi-stage builds, and zero-downtime deployment pipelines using Jenkins & GitHub Actions.

Observability & Monitoring

Integrated Prometheus time-series metrics collection and Grafana real-time dashboards for proactive alerting and diagnostic tracking.

Organization Profile & Role Definition

SystemaOps (systemaops.com)

• Specialized Cloud & DevOps Consulting Enterprise

• Empowers organizations to transition to cloud-native microservices

• Expertise in AWS, Infrastructure as Code, Kubernetes & DevSecOps

• Focus on zero-downtime deployments, cost optimization & security compliance

Role: Cloud & DevOps Engineering Intern

• Architected declarative Terraform modules for AWS VPC and ALB infrastructure

• Containerized full-stack services (Node.js, Express, React, MongoDB)

• Configured multi-stage CI/CD pipelines in Jenkins and GitHub Actions

• Implemented Kubernetes deployment manifests and Nginx Reverse Proxy

• Deployed Prometheus & Grafana stack for live metric analytics

Problem Statement & Strategic Objectives

Identified Challenges

❌ Environmental Drift: Differences between dev, staging, and production environments.

❌ Manual Provisioning: Slow, error-prone manual cloud configuration.

❌ Deployment Downtime: Risk of service interruptions during release cycles.

❌ Lack of Visibility: Insufficient real-time insights into system health and latency.

Project Objectives

✅ Infrastructure as Code: Automate 100% AWS cloud provisioning using Terraform.

✅ Containerization: Standardize microservices with Docker & Compose.

✅ CI/CD Automation: Zero-downtime deployments via Jenkins & GitHub Actions.

✅ Real-Time Monitoring: Prometheus & Grafana metrics collection & alerts.

Multi-AZ AWS 3-Tier System Architecture

Tier 1: Public Presentation Tier

AWS Application Load Balancer (ALB) • Public Subnets Multi-AZ • Nginx Reverse Proxy • SSL/TLS Termination

Tier 2: Private Application Tier

EC2 Auto-Scaling Group • Private Subnets • Kubernetes Worker Pods • Outbound via NAT Gateway

Tier 3: Isolated Database Tier

Amazon RDS (Multi-AZ MongoDB/PostgreSQL) • Private Database Subnets • Restricted Security Groups

Technologies & DevOps Toolstack

AWS Cloud

EC2, ALB, VPC, S3, RDS, CloudWatch

Terraform

Infrastructure as Code & State Management

Docker

Containerization & Multi-Stage Builds

Kubernetes

Pod Orchestration & Rolling Updates

Jenkins

CI/CD Pipeline Automation

Prometheus

Time-Series Metric Scraping & Alerts

Automated CI/CD Pipeline Workflow

Step 1	Step 2	Step 3	Step 4	Step 5
1. Code Commit	**2. Automated Test**	**3. Docker Build**	**4. Push to ECR**	**5. K8s Deploy**
GitHub Push / PR	Jest & Lint Scans	Multi-stage Image	Amazon Registry	Zero-Downtime Rollout

Observability, Monitoring & Security

Prometheus & Grafana Observability

- 📊Time-series metric collection with Prometheus scraper jobs
- 📊Node Exporter & kube-state-metrics integration
- 📊Real-time Grafana dashboards visualizing CPU, memory & request latency
- 📊Automated Alertmanager triggers for threshold violations (HTTP 5xx spikes)

Cloud Security & Best Practices

- 🔒AWS IAM Roles with Least Privilege access policy
- 🔒Private VPC Subnets for DB isolation (zero public IP exposure)
- 🔒Nginx SSL/TLS Termination & Security Hardening
- 🔒JWT Token Authentication & Role-Based Access Control (RBAC)

Key Achievements & Quantified Outcomes

100%

Infrastructure Automation

Provisioning time cut from hours to < 8 mins via Terraform

99.99%

Service Availability

Zero-downtime rolling updates managed via Kubernetes

50%+

Reduction in MTTR

Prometheus & Grafana alerting accelerated root-cause detection

CONCLUSION & LEARNING OUTCOMES

- ✓ Mastered Industry-Standard Cloud & DevOps engineering workflows at SystemaOps.
- ✓ Successfully delivered production-ready Infrastructure as Code, CI/CD, and Observability.
- ✓ Significantly enhanced technical expertise in AWS, Terraform, Docker, Kubernetes, Jenkins & Monitoring.
- ✓ Prepared public GitHub repository containing all artifacts per IILM University guidelines.

Thank You! Questions & Discussion